

- ▶ Sculptors such as Mudbox and Zbrush
- ▶ Adobe Photoshop/ Illustrator for creating textures



Skining

Animation and special effects

All movements in a game world are manifestations of some or the other kind of animation.

The animator makes the world look real and brings the characters to life. Animations can be effects, animation sequences or programmed actions comprising of graphic programming.

During production phase, the animators animate the 2D designed characters or 3D models to achieve the required effects in the game. The main concepts involved in this process are:

Kinematics

Kinematics is the study of movement of things when forces act on them. In the animation of a rig, it involves figuring out the angles of each joint as it is moved. There are two approaches to this:

Forward Kinematics (FK): With FK, we work directly with the angles of the joints in order to achieve a specific pose. Each joint is moved individually, starting with the base joints towards ones that are lower in the hierarchy. Thus when the parent joints are rotated or moved, all the child joints move as well. You must have an FK rig before you create an IK system.

Inverse Kinematics (IK): It says “If my leg is here, what should be the angle of joints”. IK lets you move the last joint in the hierarchy and have all the others follow it.

In order to achieve good, natural movement, you’ll use both of them.

Key framing

If still images are sequenced together in quick succession, human mind perceives them as continuous motion and these individual images are called frames. Typical frame rates vary from 24 to 30 frames per second (fps) depending on the format. The computer renders each frame based on